Home Automation
A setup whereby all the electronic appliances and fixtures in the house are intuitively controlled by the movement of the people through the homes. Look at the leading research at smarthome.com

Remote control is green
Controlling appliances remotely and monitoring electricity usage can decrease the carbon footprint tremendously. Such measures would also help individuals save up on running costs

of PVC – polyvinyl chloride. uPVC is unplasticised PVC, sometimes called rigid PVC. This also has good thermal insulation properties as well as sound insulation. However, not everybody would agree that these are otherwise eco-friendly – there are chemicals involved in their manufacture that have been recognised as carcinogenic.

As much recycled material as possible is used in the manufacture of the buildings, and also in their operation. For example, water drained from the sinks is recycled and used for flushing toilets and watering the small gardens (just a few square yards each) that are attached on the balconies of each flat. Non-toxic paints are also used in the buildings to help provide more hygienic conditions.

Another key element is waste management. Every kitchen sink has a waste crusher so that biological waste can be pulverised and flushed down. All general waste is also segregated for recycling as appropriate – bio-waste, solids such as tins and paper, and glass. These are then sent to private companies for recycling.

Power is also saved by having certain lighting systems automated. For example, the lights in the corridors by the lifts are usually off, but sensors detect when somebody comes into the area and turns the lights on for as long as they remain. A similar feature is also applied inside the homes where appropriate. Naturally, there are multiple backup power generators and voltage stabilisers

Just as I was about to leave J. K. Jain's office, I noticed a book about trees on his desk. He then started to tell me about which trees use up the most ground water, which use little, which attract termites, and so on. He has been researching these issues in order to determine which are the most appropriate and environmentally beneficial trees to have in the grounds of the buildings. One may not agree with many of the choices made in the style of these homes, but there is no denying that considerable attention to detail has gone into the design.

Electronically operated
What should one expect from an electronically operated home, one that is wired for modern technology? Certainly the ability to have an internet connection, but what about connectivity within the home? We might well want to connect one or two PCs, the odd laptop, printer, scanner, media hub (such as recently released by Cisco), television, audio system and so on.

Some of this will be done using USB, but the first thing I looked for in the eHomes was the presence of Ethernet connectors – I expected at least one in every room, perhaps more in the key rooms such as living room. These are completely non-existent, and this strikes me as the biggest mistake that Designarch has made. How can you possibly call an apartment an eHome, without in-built connectivity between rooms? It just does not figure.

In answer to this question, J. K. Jain told me that he would expect an owner to set up a standard broadband connection using the telephone line, and then install a WiFi router to communicate with and between devices in the home. This is all very well for those devices to communicate with the internet, but not much more than that. If I am in the living room with my laptop, can I print out on my printer connected to the PC in the spare bedroom? There are some ways in which one can stream data between devices talking to the same WiFi router, but this is not straightforward or common. At the very least, it is very much easier with Ethernet. Also, WiFi degrades very badly as you move away from the router. At the time of writing we are testing such routers here in our offices, and with a couple of concrete walls between the router and a laptop, we have been measuring something around a 60 per cent drop in signal strength and similar drop in data throughput rates. Some routers attempt to self-adjust when they receive a weak signal from a device, but this is not at all a perfect fix, and the devices themselves do not necessarily do the same. There is nothing better yet than a simple direct connection.

The central focus of the electronic control of an eHome is basically a PC with a flat touch-screen placed on the wall relatively near to the front door. This control system can be accessed also by means of a remote control - similar to one that you would use with a TV, also remotely by telephone, and, most interestingly, from a browser across the internet. This means that you can have access to all the main controls of the home when you are travelling.

The system is a normal PC running Windows, and so, at least theoretically, it could be integrated with any other PC systems in the house, although I have not seen this in operation. The control system allows access to many basic features in the home. You can turn certain appliances on and off, such as water heater, television, lighting and air conditioners, lock and open the front door, and so on. When I first discussed these ideas with Designarch I was told that only a simple on/off control was possible. However, now further levels of control are becoming available, such as one might want for turning on an oven and setting it to a particular temperature. I am not sure such an oven is available, but this is a good example of the kind of control one might need.

Some presets are available for so-called "mood settings". You can, for example, save certain lighting levels in particular rooms